

Single bin QPAD (SQPAD) approach for robust analysis of point-count data with detection error

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ABSTRACT

Bird population monitoring is often conducted using point-count surveys. Accounting for detection errors is a major challenge in analyzing these data. The commonly used methods for correcting detection errors in counts of organisms are distance sampling, removal sampling, N-mixture, and QPAD. These methods rely on multiple surveys or subdivisions within surveys (time/distance bins). The reliability of these approaches depends on the accurate estimation of distance, correct identification of individuals, and the closed population assumption. Errors in distance estimation, double counting, and mortality and migration of individuals within and between survey periods can lead to substantial biases in population density estimation. Furthermore, tracking individuals and estimating distances can be difficult in field conditions. We propose a simple modification of the QPAD method so that field observers are required to collect information only about either the occupancy status or count of individuals within a specified time interval and a specified spatial buffer, a “single bin,” around the observer’s location. We show that population density parameters are identifiable by changing the time interval and the radius of the spatial buffer for each survey location. We show that this variable-effort survey method is robust against errors in distance estimation and double counting. We also show that data collected under current protocols in North America can be analyzed using single bin QPAD. We illustrate our methodology with biologically realistic simulations and a reanalysis of some field data.

Keywords: catch per unit effort, closed population, distance sampling, effective detection radius, identifiability, N-mixture, removal sampling, RSPF condition

How to Cite

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LAY SUMMARY

- Not all birds that are present will be detected by an observer. Hence the number of birds recorded during a survey is likely to be smaller than the number that were present.
- Current methods to account for detection errors require accurately tracking individuals and their distances from the observer. This information is difficult to collect in the field.
- Double counting and errors in distance estimation can lead to biased conclusions about bird populations. These errors are exacerbated when several species are being monitored simultaneously.
- Alternative methods that do not require individual identification and distance to the observer rely on multiple visits to the same location that are logistically and financially demanding.
- We propose a new method called single bin QPAD (SQPAD) that accounts for detection error without requiring information on the exact distance to the observer, individual identification, or multiple visits. This approach simplifies the field logistics for bird monitoring programs.
- We demonstrate that SQPAD is a valid estimation method that is also robust against double counting of individuals and incorrect distance estimation.

Approche QPAD à compartiment unique (SQPAD) pour une analyse robuste des données de points d'écoute avec des erreurs de détection

RÉSUMÉ

Le suivi des populations d'oiseaux est souvent effectué à l'aide d'inventaires par points d'écoute. La prise en compte des erreurs de détection constitue un défi majeur dans l'analyse de ces données. Les méthodes couramment utilisées pour corriger les erreurs de détection dans les décomptes d'organismes sont l'échantillonnage basé sur la distance, l'échantillonnage par prélèvements, le N-mixture et le QPAD. Ces méthodes reposent sur des inventaires multiples ou des subdivisions dans les inventaires (compartiments de temps/distance). La fiabilité de ces approches dépend de la précision de l'estimation de la distance, de la bonne identification des individus et de l'hypothèse d'une population fermée. Les erreurs d'estimation de la distance, le double comptage et la mortalité et la migration des individus au cours d'une période d'inventaire et entre les périodes peuvent entraîner des biais importants dans l'estimation de la densité de la population. De plus, le suivi des individus et l'estimation des distances peuvent s'avérer difficiles sur le terrain. Nous proposons une simple modification de la méthode QPAD afin que les observateurs au terrain ne soient tenus de collecter que des informations sur le statut d'occupation ou le décompte d'individus pendant un intervalle de temps spécifié et une zone spatiale spécifiée, un « compartiment unique » (*single bin*), autour de l'emplacement de l'observateur. Nous montrons que les paramètres de densité de population sont identifiables en modifiant l'intervalle de temps et le rayon de la zone spatiale pour chaque site d'inventaire. Nous montrons que cette méthode d'inventaire à effort variable est robuste face aux erreurs d'estimation de la distance et au double comptage. Nous montrons aussi que les données collectées selon les protocoles en vigueur en Amérique du Nord peuvent être analysées à l'aide de SQPAD. Nous illustrons notre méthodologie avec des simulations biologiquement réalistes et une réanalyse de certaines données de terrain.

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Team 2024). The following describes the simulation of bird locations within a single survey location. This can be repeated over several survey locations to mimic the field data. Let us consider the landscape to be a rectangle with sides b_X and b_Y . Let D be the density and $A = b_X b_Y$ be the area of the survey location.

- Step 1: Generate a random number from $\text{Poisson}(DA)$. Let this be denoted by N .
- Step 2: Generate N random numbers from $\text{Unif}(-b_X/2, b_X/2)$ for the X -coordinates and another set of N random numbers from $\text{Unif}(-b_Y/2, b_Y/2)$ for the Y -coordinates, with coordinate $(0, 0)$ being the center of the rectangle. Then (X_i, Y_i) , $i = 1, 2, \dots, N$ are the locations of the N individual birds within the survey location. Note that these X and Y values are not related to the covariates and responses described in the notation. These simply refer to the horizontal and vertical axes.

This spatial distribution of the bird locations corresponds to the assumption that bird locations follow a HPP with intensity D . We assume that the observer is at the center of the landscape $(0, 0)$. Notice that birds are located at different distances from the observer.

Mechanism for Generating Cues

It is assumed that every individual bird emits cues over time. We do not differentiate between audible and visual cues. Time between cues is assumed to be independent of when the last cue is emitted. We also assume that the time between cues follows an exponential distribution with rate parameter δ . Thus, if we monitor an individual bird for a survey duration of, say, T , we can plot the time history of the cues. A time history of cues of hypothetical survey location with five individuals is shown in Figure 1.

In statistical parlance, this process is called a “Marked Point Process” where a “Mark” corresponds to the individual identification, and the ‘Points’ are the times at which that individual emitted a cue. We can modify this simple mechanism to incorporate possible dependence of cue rate on the season, time of the day, and other environmental conditions. Similarly, we can allow the change in the density from one location to another depending on the environmental covariates.

What Do We Observe?

Unfortunately, in field situations, our observations are not identical to Figure 1. What the observer can observe is the “perceived” version of this ideal situation. The perceived process is shown along the horizontal axis. For example, individual 2 was not detected because it did not emit a cue during the survey duration. On the other hand, individual 4 emitted a cue but was not perceived by the observer, possibly due to the distance. Individuals 1 and 3 were perceived at the time of the first cue they emitted whereas individual 5 was detected but at the second cue, not the first one. We missed the first cue from individual 5 possibly because of the distance. The third cue from individual 5 is ignored because we have already counted the individual 5 and hence “removed” from consideration. Overall, we detected 3 individuals from the 5 individuals that were present.

Hence the perceived process is quite different from the actual process because of the lack of cues and effect of distance from the observer. The challenge is to use the perceived process to infer the underlying true process, in particular, the bird density D .

Now let us mimic the data collection process according to various monitoring methods.

- 1) The complete-information method, based only on the solid dots, would declare that there are 3 individuals and notes both the distance and time information, namely, $(t_1, d_1), (t_3, d_3), (t_5, d_5)$.
- 2) Distance sampling notes that 3 individuals were observed at distances d_1, d_3 , and d_5 , respectively. It ignores the time of the cue. These are the projections of the solid dots on the distance axis (Figure 1). Notice that we keep the information on the identity of the individuals. In practice, instead of the exact distances, one may only note the distance interval in which the observation belonged to. There should be at least two pre-defined distance intervals.
- 3) Removal sampling also notes that 3 individuals were observed but collects information on the time of the first ‘perceived’ cue only at t_1, t_3 , and t_5 , respectively. These are the projections of the solid dots on the time axis (Figure 1). Notice that we keep the information on the identity of the individuals. In practice, instead of the exact times, one may only note the time interval in which the observation belonged to. There should be at least two pre-defined time intervals.
- 4) SQPAD declares that 3 individuals were observed during the survey duration $(0, t)$ and within a spatial buffer of radius r . It does not collect information about individual cue times or distance from the observer.

We collect similar information at all survey locations. This is the kind of dataset we usually have for estimating the bird density. Obviously, this is a description of a simple situation that does not involve effects of covariates on the density, cue rate or visibility. Those complications, however, can be accommodated.

Now we compare statistical properties of the estimators of bird density using distance sampling, removal sampling and SQPAD with the estimator that uses the complete information. The likelihood function for the complete-information data is described in the Supplementary Material. We consider four different situations: low density ($D = 0.5$) and low cue rate ($\delta = 0.25/\text{min}$), low density ($D = 0.5$) and high cue rate ($\delta = 0.5/\text{min}$), high density ($D = 1$) and low cue rate ($\delta = 0.25/\text{min}$), and high density ($D = 1$) and high cue rate ($\delta = 0.5/\text{min}$). The parameter $\varphi = 100\text{ m}$ was the same for all the cases. The number of survey locations (S) was 200. We generated 2,000 independent datasets and we sampled from it without replacement.

Recall that monitored data are different than the complete-information data. We used different-sized spatial buffers and different survey durations for each survey location. The maximum time duration ($T_{\max}$) was 10 min and the maximum spatial radius ($R_{\max}$) was 200 m. The minimum amount of time spent ($T_{\min}$) was 2 min. Thus, duration t_s for the s th location was one of $\{2, 4, 6, 8, 10\}$. The minimum spatial radius ($R_{\min}$) was 20 m and spatial buffer radius, r_s for the s th location was one of $\{20, 40, 60, \dots, 200\}$.

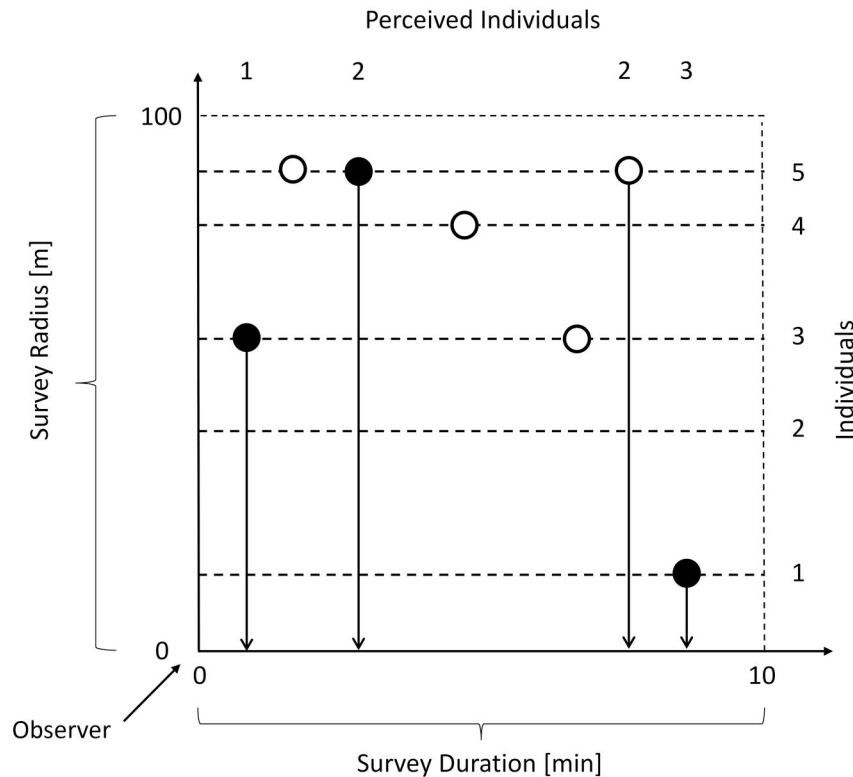


FIGURE 1. The observation process for a 10-min 100-m radius hypothetical point-count survey location where there are 5 individuals within the survey area. Circles indicate the time and distance of bird individuals giving a cue, vertical arrows pointing toward the observer indicate cues perceived by the observer, and filled dots are cues perceived first for each individual. This figure depicts an “ideal” scenario, where individuals do not move (flat lines) during the survey, there is no distance estimation error and double counting; as a result, 4 cues are detected that belong to 3 individuals. The horizontal axis is the time axis and the vertical axis is the distance axis. A cue emitted from an individual is denoted by a circle, filled (cue detected) and hollow (cue not detected). Each individual’s cue history can be deciphered along with the distance from this figure.

The detection probability was based on the Half-Gaussian distance effect model with $\varphi = 100$ m. Thus, the probability of detecting an individual at a distance r is $p(t, r; \delta_0) = 1 - \exp(-\delta_0 \exp(-(r/\varphi)^2)t)$. The Half-Gaussian model has a shoulder suggesting a slower drop in the perception of the cues near the observer than for the exponential model. This model seems to represent the reality better than the exponential model. These data can be analyzed using SQPAD method as well as all the traditional methods of distance sampling, removal sampling, and the QPAD.

For the QPAD analysis (Sólymos et al. 2013) as well as distance sampling (QAD) and removal sampling (PAD), we constructed the distance bins for a particular survey location by dividing its spatial radius r_s equally in 3 parts and time bins by dividing its survey duration t_s equally in 3 parts. To apply QPAD, we need to bin the distance-time data as described in the Field Data Analysis section. Bin sizes are different at different survey locations because the survey duration and survey buffer are different at every location. We used multinomial distance sampling and removal sampling approaches as reviewed in Sólymos et al. (2013). This analysis was conducted using the *detect* R package (Sólymos et al. 2025).

The Naïve method, that completely ignores the detection error, corresponds to using a Poisson regression model with offset for the area of the survey location. Notice that the areas are different at each survey location. This method is included to illustrate the importance of incorporating detection error in the analysis.

Simulation Results

We start with the situation when distances are measured without error and individuals are tracked perfectly and individuals do not move during the duration of the survey (Figure 2). As expected, the Naïve method is negatively biased. The complete-information likelihood, the convolution likelihood estimator, is nearly unbiased and its sampling distribution is also nearly symmetric. In the Statistical Models section, we discussed 2 versions of SQPAD; one based on the counts and one based on the occupancy status. Despite not collecting complete information, the performance of SQPAD estimator comes reasonably close to the complete-information estimator, both in terms of bias, variance, and symmetry. SQPAD estimator based on occupancy (SQPAD-O) is surprisingly comparable to SQPAD estimator based on counts (SQPAD-C). On the other hand, distance sampling (denoted by QAD), removal sampling (denoted by PAD), and QPAD have substantial mean biases. The sampling distributions of these estimators are also highly skewed, generally leading to density estimators that are substantially larger than the true density. This figure illustrates that SQPAD works and is comparable to the complete-information estimator for the data scenarios in our simulations. We also report results on the estimation of cue rate δ and distance effect parameter φ , including biases and variances in a tabular form as part of the [Supplementary Material](#).

To study the effect of errors in distance estimation and double counting on these estimators, we introduce these errors in the data generation mechanism. See Sólymos (2023) or the description of the *bSims* R package for details on how to

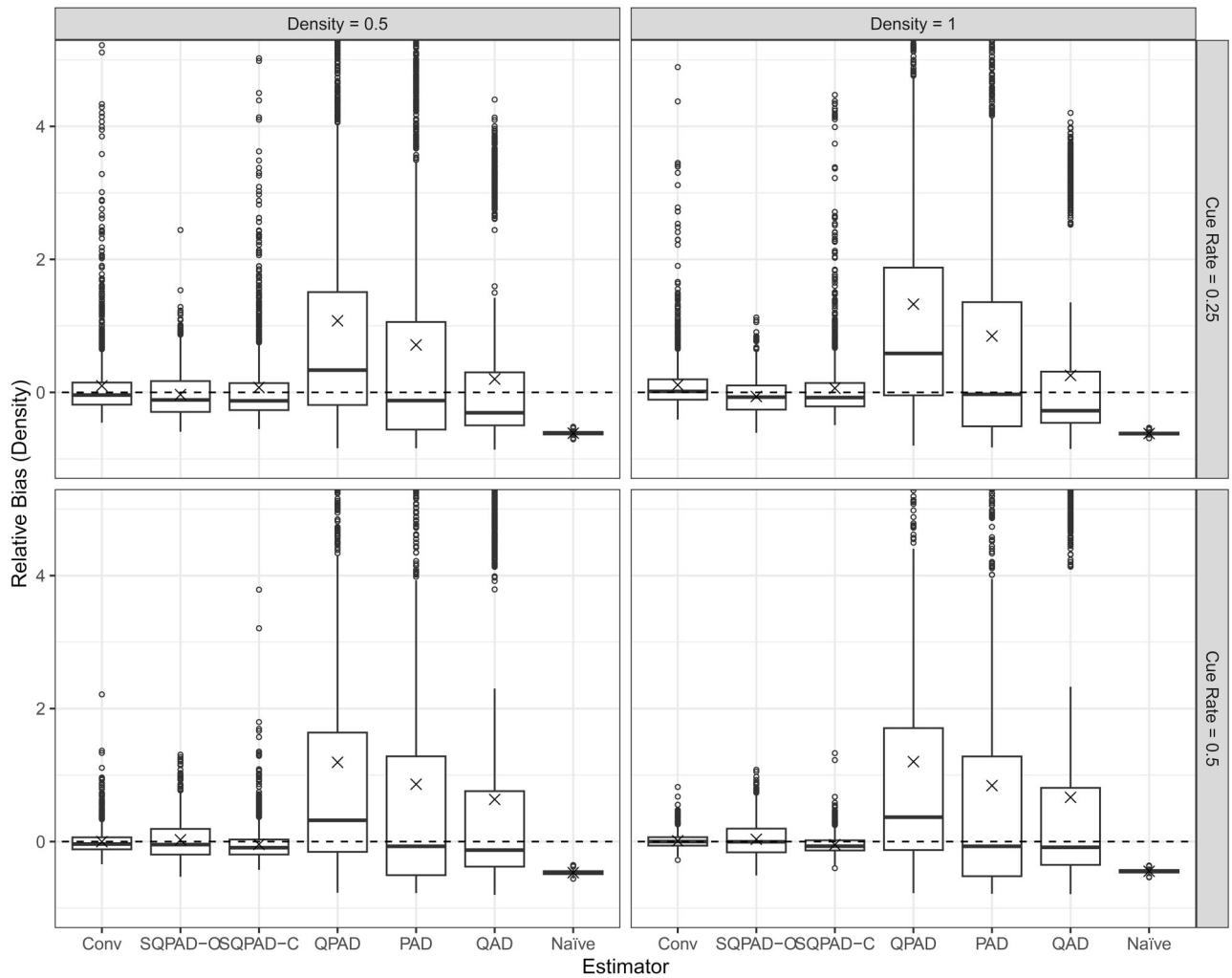


FIGURE 2. Performance of various estimators (Convolution, SQPAD-C, SQPAD-O, QPAD, distance sampling [QAD], removal sampling [PAD], and Naive) of density D . We subtract the true value of D from the estimates to represent the bias. The dotted line corresponds to zero bias. We use the Half Gaussian model for the distance effect on perception with parameter $\varphi = 100$ m. Number of survey locations is 200 and number of simulations is 2,000. We used the $0.67r$ approximation to the distance effect. Boxes and thick lines across the boxes indicate 25th, 50th, and 75th percentiles, whiskers extend to 1.5 times the inter-quartile range, open circles beyond the whiskers are outliers, x symbols indicate the mean.

introduce these errors in the data generation mechanism. It is known (Alldredge et al. 2007) that errors in distance estimation tend to shorten the true distances (Alldredge et al. 2007). To mimic these errors, we simply multiplied the true distances by 0.8 to get the “observed” distances. We introduced double counting by putting a probability of observing multiple cues emitted by the same individual as multiple individuals at 0.2. This leads to expected counts that are 1.2 times higher than the true counts. For these two scenarios, we also introduced territorial movement for the individuals according to a bivariate Normal kernel (Sóllymos 2023). Territorial movement adds variability to the distances and is also a mechanism for double counting of individuals. Figures 3 and 4 summarize these results. As expected, all estimators are biased. However, SQPAD estimators, occupancy-based and count-based, are the least biased and have smaller variability among all the estimators. This illustrates their robustness. A detailed simulation study of the coverage properties of the confidence intervals based on SQPAD estimators needs to be conducted.

In our simulations with double counting error (Figure 4), the occupancy based SQPAD estimator was more biased than the count-based approaches (convolution likelihood and SQPAD-C)

when density and cue rates were both high. This is possibly caused by having too few 0's (10–24%) in the simulated datasets. It is generally known that occupancy-based methods are less accurate when counts are high and the frequency of 0 values is low.

The QPAD estimator proposed by Sóllymos et al. (2013) uses the same information as the complete-information convolution likelihood estimator. However, it uses a somewhat untenable assumption that the probability of detection can be written as a product of two probabilities: one that depends only on the distance and the other that only depends on the time. This is what is termed as the ‘independence model’ (Sóllymos et al. 2013, Martin-Schwarze et al. 2021). This assumption allows the estimation of the cue rate and visibility bias in a simple fashion from field data collected under many different protocols (Edwards et al. 2023). However, if this assumption is not satisfied, as is the case in our simulations, the QPAD estimator of density is biased. This bias is evident in Figures 2–4. We provide information on the behavior of the estimates of cue rate and distance effect parameters in the Supplemental Material. As is the case for density estimates, SQPAD estimates are better than QPAD, distance sampling, and removal sampling estimates.

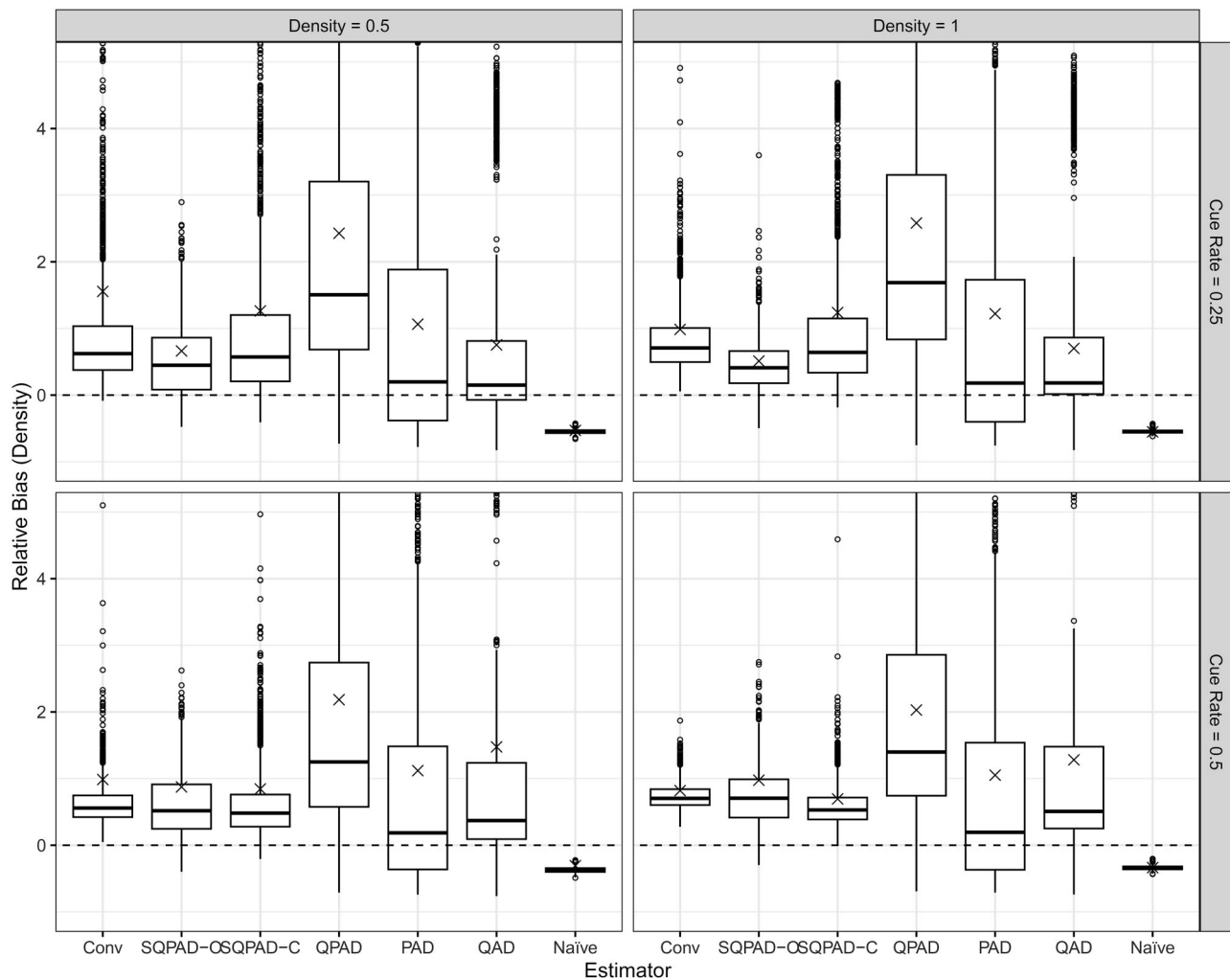


FIGURE 3. Performance of various estimators (Convolution ML, SQPAD-C, SQPAD-O, QPAD, distance sampling [QAD], removal sampling [PAD], and Naïve) of density D when distances are measured with error (20% negative bias). We subtract the true value of D from the estimates to represent the bias. The dotted line corresponds to zero bias. We use the Half Gaussian model for the distance effect on perception with parameter $\phi = 100$ m. Number of survey locations is 200 and number of simulations is 2,000. See Figure 2 legend for the explanation of box and whiskers statistics.

AN EXAMPLE FIELD DATA ANALYSIS

We reanalyzed data from [Van Wilgenburg et al. \(2017\)](#) using SQPAD. We present results only for 4 of the species that were monitored: *Spizella passerina* (Chipping Sparrow), *Junco hyemalis* (Dark-eyed Juco), *Seiurus aurocapilla* (Ovenbird), and *Catharus ustulatus* (Swainson's Thrush). These species represent different average densities and cue rates based on previous QPAD estimates ([Van Wilgenburg et al. 2017](#)). Code for SQPAD analysis and results for the other 8 species can be found in the [Supplementary Material](#).

The data consisted of 351 point-count surveys performed by human observers. We used latitude (centered and scaled by its standard deviation) as a covariate for density (X) and time since local sunrise (hours since sunrise divided by 24) and its quadratic term as covariates for the cue rate (Z). For each survey, distances to the individuals were reported to belong to 1 of 2 bins; <50 m and 50–100 m. Similarly, times to the first cue were reported to belong to 1 of 3 bins: <3 min, 3–5 min, and 5–10 min.

A hypothetical but typical datum corresponding to a single survey is a 2-way table ([Figure 5A](#)). At this survey location,

we observed 2 distinct individuals during the first 3 min and both were within 50 m of the observer. These 2 individuals are excluded from further consideration. Other cells can be interpreted similarly. We observed a total of 5 individuals within 100 m of the observer location and during the 10-min duration.

This field data was not collected according to the variable effort survey design. However, one can turn these data into data observed under the variable effort survey design. For each survey location, one can randomly select a number from (50,100) and another number from (3,5,10). Suppose for the above survey location, random numbers were 50 and 5, respectively. Then we can replace the above 2-way table with a vector $(y_s, r_s, t_s) = (3, 50, 5)$ as the observed data for that survey location.

The scheme described above of turning a typical distance-time binned dataset into the variable effort survey design dataset, however, loses information because we use one randomly selected subset from all possible subsets. The number of possible subsets is equal to the number of distance bins multiplied by the number of time bins. In the case above, there are 6 possible subsets. We can estimate density D using

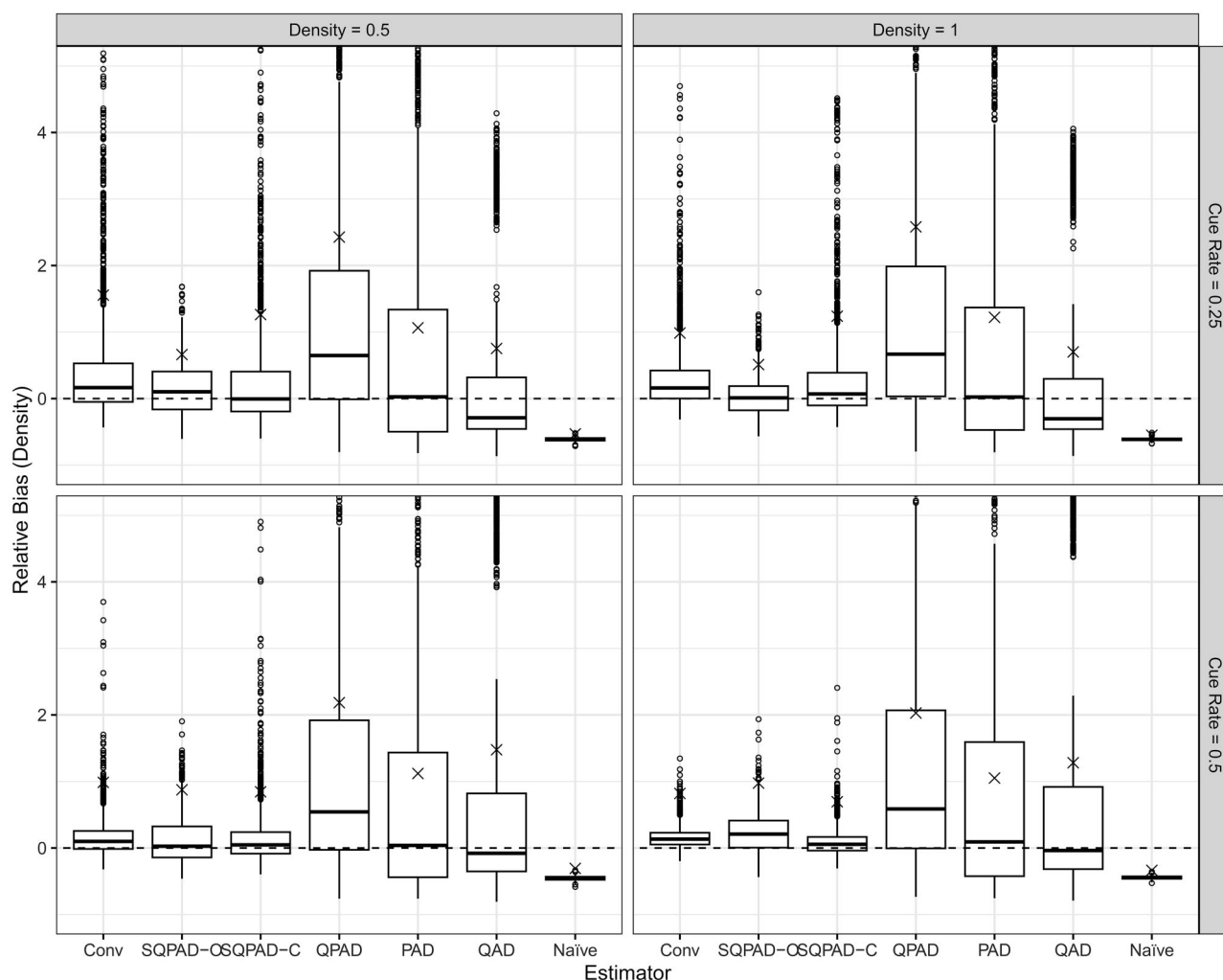


FIGURE 4. Performance of various estimators (Convolution ML, SQPAD-C, SQPAD-O, QPAD, distance sampling [QAD], removal sampling [PAD], and Naïve) of density D when there was double counting (20% positive bias). We subtract the true value of D from the estimates to represent the bias. The dotted line corresponds to zero bias. We use the Half Gaussian model for the distance effect on perception with parameter $\phi = 100$ m. Number of survey locations is 200 and number of simulations is 2,000. See Figure 2 legend for the explanation of box and whiskers statistics.

A			B		
Time/Distance	0-50 m	50-100 m	Time	Distance	Count
0-3 min	2	0	0-3 min	0-50 m	2
3-5 min	1	1	0-3 min	0-100 m	2
5-10 min	0	1	0-5 min	0-50 m	3
			0-5 min	0-100 m	4
			0-10 min	0-50 m	3
			0-10 min	0-100 m	5

FIGURE 5. Count of individuals' first detections by time and distance intervals for a hypothetical survey under the combination of removal and distance sampling required for multiple-bin QPAD (**A**). The same counts from the same survey are organized according to the cumulative time and distance intervals and the corresponding counts as required for the composite likelihood SQPAD (**B**).

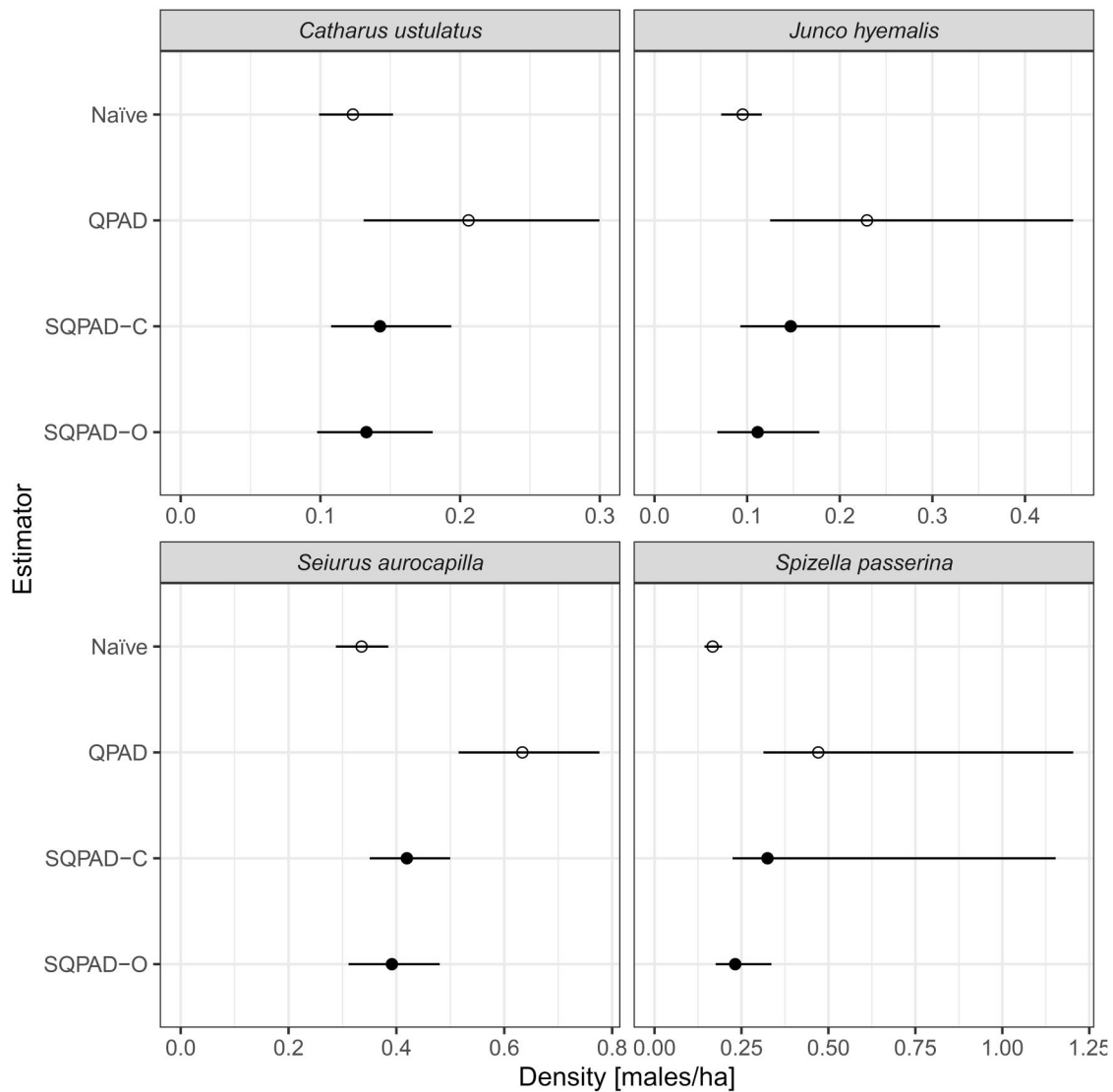


FIGURE 6. Results from the reanalysis of field data for 4 species, *Catharus ustulatus*, *Junco hyemalis*, *Seiurus aurocapilla*, and *Spizella passerina*, using SQPAD-C, SQPAD-O, QPAD, and Naïve estimators. SQPAD estimates are based on the composite likelihood approach. SQPAD estimates are uniformly lower than the corresponding QPAD estimates. We used the Half Gaussian model for the distance effect on perception.

Also, a model can be wrong in many directions and these tests cannot have good power in every such direction. We believe that one can compare the appropriateness of 2 models but model adequacy of a single model is a tricky and essentially an unsolved problem because what is an adequate fit is a decision problem; a particular Chi-square value might be reasonable for some kinds of decisions but not for others.

Model Diagnostics

The model selection procedure and goodness-of-fit tests, being omnibus tests, cannot diagnose validity of individual components of the model for density parameter and model for detection error separately. For example, to judge the appropriateness of the time-distance effect on the detection, one needs to have a complete record for every individual: the time of all the cues and the distance, at which they emitted (e.g., [Martin-Schwarze et al. 2021](#)). This kind of information, without any errors in data collection, is nearly impossible to get in practice. This is particularly true when the cues are

aural rather than visual. Hence, we doubt the practical utility of such diagnostic tests in bird monitoring.

Furthermore, in practice, distance and first arrival times data are usually collected in 2 or 3 intervals. Given such crude resolution data, the power to discriminate between different forms of distance effect models is likely to be quite moderate. On the other hand, although SQPAD samples only one bin at any location, by varying the time and distance bin sizes at a finer scale from location to location, we expect SQPAD to be more sensitive to model differences than QPAD and distance sampling. The sensitivity to differences in the detection models under these different protocols needs to be studied further.

Some Caveats

SQPAD model is a hierarchical model—a model that minimally consists of 2 components: a process model and an observation model ([Lele et al. 2007](#))—and an inherent characteristic of any hierarchical model is that one cannot

we have shown, a reasonable mechanism for detection, that also underlies the commonly used methods of removal sampling and QPAD, leads to such a model for the probability of detection. We have added no new assumptions beyond what is assumed in these current methods. The main change that SQPAD method introduces is the variation in the survey duration and survey radius from location to location.

If the variation in survey duration and survey buffer is low or if they are identical from location to location, this method will not work. This is consistent with the result that higher variation in covariate values leads to better regression model estimation (Neter et al. 1983). This was also observed in the context of regression models used in occupancy studies (von Hirschheydt et al. 2023). Misclassification of individuals as being inside the buffer when they are outside and vice versa may also lead to some bias. However, simulations that include the errors in distance estimation indicate that such misclassification may not have a substantial impact.

We derived the detection probability model based on a specific mechanistic model for the arrival and perception of cues. Although not tested using simulations, we believe, mathematically, we only need to have a detection probability model to be a function of the survey duration (time) such that, with all other covariates including distance held constant, such that the probability of detection strictly increases from 0 to 1 as the duration increases from 0 to ∞ . Any such function along with a spatial homogeneous Poisson point process for the location of individuals guarantees identifiability of the density parameter D (Sólymos et al. 2012, Sólymos and Lele 2016).

When using QPAD, we collect data in multiple time and distance bins at the same survey location. Similar to the results in Lele et al. (2012), what we have shown is that, without any additional assumptions, one can estimate bird density even if one collects data in a single time and distance bin, as long as the bin sizes vary from location to location. The *within* location changes in bin sizes for QPAD is replaced by changes in bin sizes *between* survey locations in SQPAD. This simple modification in data collection protocol allows us to collect data in a simpler fashion without losing any inferential properties of QPAD. We have also removed the troublesome time-distance independence assumption required by QPAD.

As information from autonomous recording units (ARUs) becomes more readily available (Shonfield and Bayne 2017), there is a growing need for analytical methods that work with the characteristics of data from ARUs. ARU datasets typically do not have information on individual distances and identification of individuals is also difficult or impossible, making the use of traditional methods, such as distance sampling or removal sampling difficult. On the other hand, with ARU data, the physical radius r for the spatial buffer can be reasonably replaced by the sound pressure (Yip et al. 2020) and they are also more reliable to get occupancy data than to get the counts. Hence SQPAD-O is well equipped to deal with the ARU data. Like the QPAD, it is also well-equipped to deal with data that are collected under differing protocols between different monitoring programs. In recent years, citizen science data on the occurrence of bird species has become commonly available. Most avid observers can collect occupancy data and if the individual they heard or saw was within a specified buffer. Hence, we suspect SQPAD may be able to deal with such data as well. We will be studying these extensions in a future publication.

Supplementary material

Supplementary material is available at *Ornithological Applications* online.

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Ethics statement

This study only used simulated data sets or observations taken from the literature, and no birds were captured or manipulated.

Conflict of interest statement

The authors have no conflicts of interest to declare.

Author contributions

S.L. and P.S. conceived the idea, designed and performed the simulation experiments, analyzed the data, and wrote the paper together.

Data availability

Analyses reported in this article can be reproduced using data and code archived on Zenodo (<https://doi.org/10.5281/zenodo.16172209>).

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